

• ACTIVITY.

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#	Scenario	Technique	Reason.
1.	CCTV image contains Salt-&-pepper noise	Median Filter	Removes Salt-&-pepper noise while preserving edges.
2.	Satellite image corrupted by gaussian noise	Gaussian Filter	Smooths gaussian noise effectively.
3.	Medical X-rays has random noise but edges must be preserved	Bilateral Filter	Reduces noise while preserving fine edges.
4.	Scanned document has black and white dots	Median Filter	Eliminates impulse noise from documents.
5.	Low-light image shows heavy grain	Non-Local mean (NLM) Denoising	Removes heavy noise while retaining image details.
6.	Foggy road image has very low contrast	Histogram Equalization	Improves contrast and makes objects more visible.
7.	Underwater image appears dull and washed out	Contrast Stretching	Enhances brightness & improves overall visibility.
8.	Chest X-ray has poor visibility of internal structures	CLAHE.	Enhances local contrast without over-amplifying noise.

9. Night-time traffic

Histogram
Equalization

Increases
intensity variation
& improves
visibility.

10. Grayscale image
uses only a narrow
intensity range

~~Contrast
stretching~~

Expands the
intensity range
for better image
quality.

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